

## Group Project

Learners will first complete **Phase 1: Ideation and Planning (Module 3)**. This will be followed by a live class during which learners will present their plans to **Dr. Tasneem**, who will provide direct feedback. After incorporating this feedback, learners will submit a refined version of their work in **Phase 2 (Module 4)**.

**Submission Requirements:** To be eligible for certification, all learners must submit both assignments—Phase 1 and Phase 2.

### *Module 3 - Phase 1 Ideation and Planning*

So far, we have explored the foundation of STEM: from its history and relevance in our curriculum to the core characteristics that define it, as well as the pedagogical approaches that make it effective. We also reviewed the STEM Unit Storyline template and heard valuable insights from STEM experts on the best practices for implementing STEM both in the classroom and at a school-wide level.

Now, it's time to put theory into practice! Since STEM emphasizes applying knowledge to real-world situations and fostering collaboration, this project will give you the opportunity to work together and apply what you've learned.

In the final two modules, we will focus on the **Group Project: Designing a STEM Unit Storyline**. Here, you will work in groups of 4-5 to design a functional STEM Unit Storyline that aligns with classroom learning goals.

Please keep the following guidelines in mind for your project:

- **Scope:** While full unit storylines can be extensive, aim to create a storyline with 6-8 lessons, covering approximately 2-3 weeks.
- **Real-World Connection:** Make sure the storyline you create connects with real-world applications relevant to your students.

This project will not only reinforce your understanding of STEM principles but will also provide a collaborative experience that mirrors how STEM is taught and learned effectively.

In the next module, we will dive into the detailed design of your STEM Unit Storyline. In this module, let's begin by brainstorming a **meaningful problem or challenge** that you want your students to address.

The problem or challenge you choose will form the foundation of your storyline. Please consider the following factors to ensure the challenge is impactful and aligns well with the STEM approach:

- **Alignment with Course Content:**

Select a problem that ties closely with the core Science and Math concepts you are teaching. This alignment will ensure that the storyline reinforces essential learning goals and seamlessly integrates STEM within the curriculum.

- **Contextual Relevance:**

Think about the local context—focus on a problem that reflects the unique needs, challenges, or interests of students in Pakistan. A contextually relevant problem will make learning more relatable and meaningful for your students, fostering a stronger connection to STEM.

**Template:****STEM Unit Storyline**

|                                                                                                                                                                                                                                                                                       |
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| Names of Group Members                                                                                                                                                                                                                                                                |
| Proposed Title of Unit                                                                                                                                                                                                                                                                |
| Grade Level                                                                                                                                                                                                                                                                           |
| Learning Outcomes of Unit                                                                                                                                                                                                                                                             |
| <b>Context/Problem:</b><br><i>Write a concise description of the real-world problem or challenge you'll be presenting to students. Explain why this problem is relevant to their local or national context and how it will engage them in applying both Science and Math content.</i> |
| <b>Science Content Being Used</b><br><i>Briefly list the specific Science concepts or topics that align with the problem you have chosen.</i>                                                                                                                                         |
| <b>Mathematics Content Being Used</b><br><i>Similarly, indicate the relevant Mathematics concepts or skills that will support solving the problem.</i>                                                                                                                                |
| <b>Technology and Engineering Being Used</b><br><i>Finally, share your idea for integrating technology as an outcome of Engineering Design Process.</i>                                                                                                                               |

**Group Project - Module 4****Phase II- Designing STEM Unit Storyline**

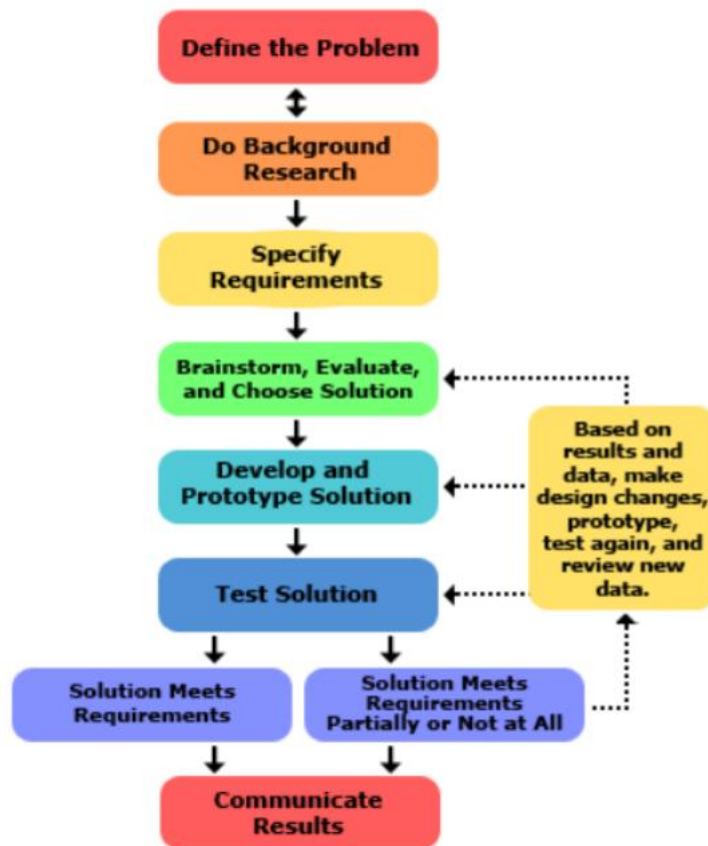
In this phase, each group will collaboratively design the STEM Unit Storyline using the knowledge from the lesson videos and your agreed-upon problem statement. This phase has 1 deliverable

**Deliverable: STEM Unit Storyline**

Each group will submit a STEM Unit Storyline based on the template provided. Your storyline should offer a clear and comprehensive structure that maps out the learning journey, integrating the core characteristics

of STEM and aligning with the Engineering Design Process (EDP). This will include lesson-specific driving questions, learning outcomes, tasks, and assessment strategies.

Please note that the total number of lessons can be between 6-8.



**Rubric****STEM Integration Assessment Tool**

Name of STEM Unit Storyline Designer \_\_\_\_\_

STEM Unit Storyline Title \_\_\_\_\_ Grade: \_\_\_\_\_

| <b>STEM Integration Criteria</b>                                                                                                                                            | <b>Excellent (5)</b>                                                                                               | <b>Satisfactory (3)</b>                                                                                   | <b>Needs Improvement (1)</b>                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>The STEM Unit Storyline...</b>                                                                                                                                           |                                                                                                                    |                                                                                                           |                                                                          |
| <b>Real World Problems</b>                                                                                                                                                  |                                                                                                                    |                                                                                                           |                                                                          |
| Used a real-world or authentic problem as a context for learning.                                                                                                           | Uses a meaningful, authentic real-world problem that drives all learning experiences in the unit.                  | Uses a real-world problem, but it is not consistently used to guide learning throughout the unit.         | Problem is unclear, artificial, or not connected to student learning.    |
| <b>Context Integration</b>                                                                                                                                                  |                                                                                                                    |                                                                                                           |                                                                          |
| Provided a motivating context with a compelling purpose ( <i>what, why, for whom, and explicit integration between the problem context and content learning outcomes</i> ). | Clearly defines the problem, purpose, audience, and explicitly links the context to all content learning outcomes. | Context is motivating but lacks clarity in purpose, audience, or explicit alignment to learning outcomes. | Context is weak, unclear, or disconnected from learning outcomes.        |
| <b>Content Integration</b>                                                                                                                                                  |                                                                                                                    |                                                                                                           |                                                                          |
| Stated learning outcomes of the overall STEM unit storyline.                                                                                                                | Learning outcomes are clearly stated, measurable, and aligned with the unit storyline.                             | Learning outcomes are stated but are broad, vague, or partially aligned.                                  | Learning outcomes are missing or unclear.                                |
| Integrated science concepts that are grade-level appropriate and in line with the NCP (2022).                                                                               | Science concepts are grade-appropriate, aligned with NCP (2022), and essential to solving the design challenge.    | Science concepts are aligned with NCP (2022) but weakly connected to the Storyline.                       | Science concepts are inaccurate, missing or not aligned with NCP (2022). |
| Integrated mathematics concepts that are grade-level appropriate and in line with the NCP (2022).                                                                           | Mathematics concepts are grade-appropriate, aligned with NCP (2022), and used meaningfully in problem-solving.     | Mathematics concepts are aligned but used procedurally or with limited relevance.                         | Mathematics integration is minimal, incorrect, or absent.                |
| Integrated technology as an outcome of the Engineering Design Process.                                                                                                      | Technology is purposefully developed or applied as a direct outcome of the Engineering Design Process.             | Technology is used but not clearly connected to engineering outcomes.                                     | Technology integration is minimal or unrelated to engineering.           |
| Included assessments closely aligned with the STEM unit's learning outcomes, providing students with varied opportunities to demonstrate their learning.                    | Includes multiple, well-aligned formative and summative assessments that allow varied demonstrations of learning.  | Assessments are present and partially aligned but limited in variety or depth.                            | Assessments are poorly aligned or missing.                               |
| <b>Centrality of Engineering</b>                                                                                                                                            |                                                                                                                    |                                                                                                           |                                                                          |
| Provided explicit opportunity to use Engineering Design Process.                                                                                                            | Students explicitly engage in multiple stages of the Engineering Design Process.                                   | Engineering Design Process is included but not fully emphasized.                                          | Engineering Design Process is implicit or absent.                        |
| Provided explicit opportunity to plan & test the prototype.                                                                                                                 | Students plan, build, and test prototypes using defined criteria and constraints.                                  | Students test prototypes but planning or                                                                  | Prototype planning and testing are minimal or missing.                   |

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|                                                                                                                                                                                                                  |                                                                                                                   | evaluation is limited.                                                         |                                                                   |
| Provided explicit opportunity to re-design the prototype.                                                                                                                                                        | Students redesign prototypes based on test data, feedback, and evidence.                                          | Redesign occurs but lacks clear evidence-based justification.                  | No redesign opportunity is provided.                              |
| <b>21st Century Skills</b>                                                                                                                                                                                       |                                                                                                                   |                                                                                |                                                                   |
| Required students to collaborate with others to demonstrate teamwork and that each member of the team is needed for the completion of the team task.                                                             | Students collaborate effectively with shared responsibility and interdependent roles.                             | Students work in groups, but collaboration is uneven or loosely structured.    | Little to no meaningful collaboration occurs.                     |
| Required students to communicate science, mathematics concepts, and engineering solutions/products ( <i>e.g., oral, written, or using visual aids such as charts or graphs using evidence-based reasoning</i> ). | Students clearly communicate ideas using evidence through oral, written, and/or visual formats.                   | Students communicate ideas but with limited clarity or evidence.               | Communication is unclear, incomplete, or unsupported by evidence. |
| <b>STEM Practices</b>                                                                                                                                                                                            |                                                                                                                   |                                                                                |                                                                   |
| Included lesson-specific driving question/s.                                                                                                                                                                     | Each lesson includes a clear, meaningful driving question aligned to the design challenge.                        | Driving questions are present but unclear or inconsistently aligned.           | Driving questions are missing or ineffective.                     |
| Engaged students in varied pedagogies, tasks, or activities ( <i>e.g., involving inquiry/ hands-on /questions/POE/ discrepant event/model and modeling, etc.</i> ).                                              | Students engage in varied, student-centered strategies ( <i>e.g., inquiry, hands-on learning, modeling</i> ).     | Some variation in pedagogy exists, but instruction is mostly teacher-directed. | Instruction relies on limited or passive teaching methods.        |
| Required students to use evidence-based reasoning ( <i>e.g., involving students to collect and analyze information or data before arriving at a solution</i> ).                                                  | Students collect, analyze, and use data to justify decisions and solutions.                                       | Students use data but with limited analysis or reasoning.                      | Evidence-based reasoning is minimal or absent.                    |
| Included explicit connections to the overall design challenge/context in every lesson so that students understand why each lesson is important.                                                                  | All lessons clearly connect to the overall design challenge and purpose.                                          | Connections are present but inconsistently reinforced.                         | Lessons are disconnected from the design challenge.               |
| Required students to apply technology tools (digital and non-digital) for content-specific learning, research, information analysis, and problem-solving, communication, collaboration, and/or decision-making.  | Students appropriately use technology tools for learning, research, analysis, collaboration, and problem-solving. | Technology tools are used but with limited purpose or depth.                   | Technology use is minimal or ineffective.                         |
| Included resources/materials needed for each lesson.                                                                                                                                                             | All lessons clearly list appropriate resources and materials.                                                     | Resources are listed but incomplete or unclear.                                | Resources and materials are not specified.                        |
| <b>Promoting STEM careers</b>                                                                                                                                                                                    |                                                                                                                   |                                                                                |                                                                   |
| Required students to get introduced to any STEM career or engage with STEM professionals.                                                                                                                        | Students meaningfully explore STEM careers or interact with STEM professionals connected to the unit.             | STEM careers are mentioned but not deeply integrated.                          | No STEM career connections are included.                          |

**Any other comments:**

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